The Complete Treatise on Solving a System of 3 Linear Equations in 3 Unknowns

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Abstract

A system of three linear equations in three unknowns is one of the simplest nontrivial models of simultaneous relationships. Although substitution, elimination, Gaussian elimination, matrix inversion, and Cramer's rule are often presented as separate techniques, they are manifestations of a common linear-algebraic structure. This paper develops that structure and compares the principal exact and numerical methods, including Gaussian and Gauss–Jordan elimination, matrix inversion, Cramer's rule, LU and QR factorization, and iterative numerical methods.

Applications are discussed across mathematics, statistics, economics, finance, physics, chemistry, biology, psychology, psychiatry, philosophy, political science, geopolitics, diplomacy, international relations, welfare economics, warfare economics, computer science, algorithms, data science, and artificial intelligence. The applications are understood as mathematical modeling examples: a linear system supplies a framework for representing simultaneous constraints, while the substantive validity of a model depends on empirical assumptions, measurement, causality, and domain-specific evidence.

The treatise ends with "The End"

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1 Introduction

Consider the system

$$\begin{aligned} a_{11}x + a_{12}y + a_{13}z &= b_1, \\ a_{21}x + a_{22}y + a_{23}z &= b_2, \\ a_{31}x + a_{32}y + a_{33}z &= b_3. \end{aligned}$$

There are three equations and three unknowns, namely x, y, and z. The central question is whether there exists a triple (x, y, z) satisfying all three equations simultaneously.

In elementary algebra, one may approach the problem by substitution or elimination. In linear algebra, the same system is written compactly as

$$A\mathbf{x} = \mathbf{b},$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

The determinant

$$D = \det(A)$$

plays a central role. If $D \neq 0$, then A is invertible and there is exactly one solution,

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

If $D = 0$, uniqueness fails: the system either has no solution or infinitely many solutions [3, 2, 1].

2 The Geometric Meaning

Each linear equation in three variables represents a plane in three-dimensional Euclidean space. Therefore a 3×3 linear system asks for the common intersection of three planes.

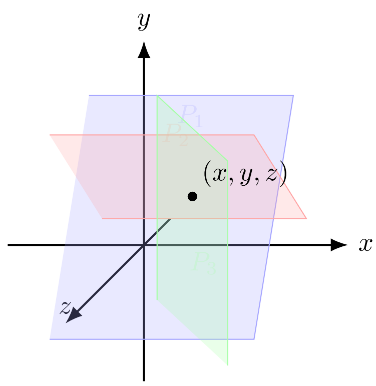

Figure 1: The geometric meaning.

For a generic nonsingular system, the three planes intersect at one point. Special configurations produce either a line, a plane, or no common intersection. This geometric interpretation explains the algebraic classification of solutions [2, 3].

3 Method I: Substitution

Substitution is the most direct algebraic method.

Suppose

$$\begin{aligned} x + y + z &= 6, \\ 2x - y + z &= 3, \\ x + 2y - z &= 2. \end{aligned}$$

The first equation gives

$$x = 6 - y - z.$$

Substitution into the second equation gives

$$2(6 - y - z) - y + z = 3,$$

and hence

$$3y + z = 9.$$

Substitution into the third equation gives

$$6 - y - z + 2y - z = 2,$$

so

$$y - 2z = -4.$$

The original 3×3 problem has therefore become a 2×2 problem:

$$\begin{aligned} 3y + z &= 9, \\ y - 2z &= -4. \end{aligned}$$

Solving gives

$$y = 2, \qquad z = 3,$$

and consequently

$$x = 1.$$

Thus

$$\boxed{(x, y, z) = (1, 2, 3)}.$$

Substitution is conceptually transparent, although expression growth can make it inefficient for larger systems.

4 Method II: Elimination

Elimination systematically removes variables by taking linear combinations of equations.

For the same system,

$$\begin{aligned} x + y + z &= 6, \\ 2x - y + z &= 3, \\ x + 2y - z &= 2, \end{aligned}$$

replace the second equation by

$$E_2 - 2E_1,$$

and the third by

$$E_3 - E_1.$$

This gives

$$\begin{aligned} x + y + z &= 6, \\ -3y - z &= -9, \\ y - 2z &= -4. \end{aligned}$$

The resulting equations contain fewer variables. Repeating the process produces a triangular system, which can be solved by back-substitution.

Elimination is the conceptual ancestor of Gaussian elimination [3, 6].

5 Method III: Gaussian Elimination

Gaussian elimination converts the augmented matrix into row-echelon form.

The system above corresponds to

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 2 \end{array}\right].$$

Elementary row operations are

$$R_i \leftarrow R_i + cR_j, \qquad R_i \leftrightarrow R_j, \qquad R_i \leftarrow cR_i, \quad c \neq 0.$$

They preserve the solution set.

The elimination procedure can be expressed as follows.

Pseudocode

```
GaussianElimination(A, b):
    form augmented matrix M = [A | b]

    for k = 1, ..., n-1:
        choose a suitable pivot in column k
        interchange rows if necessary

        for i = k+1, ..., n:
            m = M[i,k] / M[k,k]
            replace row i by row i - m(row k)

    solve the resulting triangular system by back-substitution
    return x
```

For a dense $n \times n$ matrix, Gaussian elimination requires approximately

$$\frac{2}{3}n^3$$

floating-point operations, followed by $O(n^2)$ work for back-substitution [4, 6].

For $n = 3$, the arithmetic is small enough that almost any method is computationally feasible. For large systems, however, algorithmic complexity and numerical stability become decisive.

6 Method IV: Gauss–Jordan Elimination

Gauss–Jordan elimination continues elimination above the pivots as well as below them. The target is

$$[A \mid \mathbf{b}] \longrightarrow [I \mid \mathbf{x}].$$

Thus, after reduction,

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & x \\ 0 & 1 & 0 & y \\ 0 & 0 & 1 & z \end{array}\right]$$

directly displays the solution.

Gauss–Jordan elimination is particularly useful for teaching, symbolic manipulation, and computing inverses, although ordinary Gaussian elimination is generally preferable for numerical solution of a single system [3, 2].

7 Method V: Matrix Inversion

Write the system as

$$A\mathbf{x} = \mathbf{b}.$$

If

$$\det(A) \neq 0,$$

then A^{-1} exists. Multiplying by A^{-1} gives

$$A^{-1}A\mathbf{x} = A^{-1}\mathbf{b},$$

and because

$$A^{-1}A = I,$$

we obtain

$$\boxed{\mathbf{x} = A^{-1}\mathbf{b}}.$$

The method is mathematically elegant but should not normally be interpreted as "calculate the inverse first" when numerical efficiency is the goal. Numerical linear algebra generally recommends factorization methods, particularly LU factorization, instead of explicitly forming an inverse [4, 5].

8 Method VI: Cramer's Rule

Let

$$D = \det(A).$$

For

$$A\mathbf{x} = \mathbf{b},$$

Cramer's rule states

$$x = \frac{D_x}{D}, \qquad y = \frac{D_y}{D}, \qquad z = \frac{D_z}{D},$$

where D_x is obtained by replacing the first column of A by $\mathbf{b}$, and similarly for D_y and D_z.
For

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

the determinant is

$$D = a(ei - fh) - b(di - fg) + c(dh - eg).$$

Cramer's rule is useful for theoretical derivations and small systems, but it is generally inefficient for large numerical problems [1, 2].

9 Why Cramer's Rule Works

One proof follows from the multilinearity and alternating properties of the determinant.

Suppose

$$A = [\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3],$$

and

$$\mathbf{b} = x\mathbf{a}_1 + y\mathbf{a}_2 + z\mathbf{a}_3.$$

Replacing the first column by $\mathbf{b}$ gives

$$D_x = \det[\mathbf{b}\ \mathbf{a}_2\ \mathbf{a}_3].$$

By linearity,

$$D_x = x\det[\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3] + y\det[\mathbf{a}_2\ \mathbf{a}_2\ \mathbf{a}_3] + z\det[\mathbf{a}_3\ \mathbf{a}_2\ \mathbf{a}_3].$$

The last two determinants vanish because they contain repeated columns. Therefore

$$D_x = xD.$$

If $D \neq 0$,

$$x = \frac{D_x}{D}.$$

The same argument establishes the formulas for y and z [1].

10 Method VII: LU Factorization

An invertible matrix can often be decomposed as

$$A = LU,$$

where L is lower triangular and U is upper triangular.

Then

$$A\mathbf{x} = \mathbf{b}$$

becomes

$$LU\mathbf{x} = \mathbf{b}.$$

Introduce

$$L\mathbf{y} = \mathbf{b},$$

followed by

$$U\mathbf{x} = \mathbf{y}.$$

Both systems are triangular and can therefore be solved efficiently by forward and backward substitution.

LU factorization is especially valuable when the same coefficient matrix is used with many different right-hand sides:

$$A\mathbf{x}_1 = \mathbf{b}_1, \quad A\mathbf{x}_2 = \mathbf{b}_2, \quad \ldots$$

The factorization of A can be reused [4, 6].

11 Method VIII: QR Factorization

Another factorization is

$$A = QR,$$

where Q is orthogonal and R is upper triangular.
Since

$$Q^\mathsf{T} Q = I,$$

we obtain

$$A\mathbf{x} = \mathbf{b} \Longrightarrow QR\mathbf{x} = \mathbf{b}.$$

Multiplying by Q^T gives

$$R\mathbf{x} = Q^\mathsf{T}\mathbf{b}.$$

QR factorization is particularly important for least-squares problems and numerically stable computation [5, 4].

12 Classification of Solutions

The rank of the coefficient matrix and the augmented matrix provides a general classification.
Let

$$A\mathbf{x} = \mathbf{b}.$$

Then:

- If $\operatorname{rank}(A) = \operatorname{rank}([A \mid \mathbf{b}]) = 3$, there is exactly one solution.
- If $\operatorname{rank}(A) = \operatorname{rank}([A \mid \mathbf{b}]) < 3$, there are infinitely many solutions.
- If $\operatorname{rank}(A) < \operatorname{rank}([A \mid \mathbf{b}])$, there is no solution.

This is the Rouché–Capelli theorem [3, 2].
For a 3×3 matrix,

$$\det(A) \neq 0 \iff \operatorname{rank}(A) = 3 \iff A^{-1} \text{ exists.}$$

Consequently,

$$\boxed{\det(A) \neq 0 \Longrightarrow \text{exactly one solution.}}$$

13 Comparison of the Principal Methods

Method	**Principal idea**	**Typical use**
Substitution	Solve and substitute	Small hand calculations
Elimination	Linear combinations	Hand calculations
Gaussian	Row-echelon form	General numerical solution
Gauss–Jordan	Reduced row-echelon form	Teaching/inverses
Matrix inverse	$\mathbf{x} = A^{-1}\mathbf{b}$	Theory and symbolic work
Cramer's rule	Determinant ratios	Small exact systems
LU	$A = LU$	Repeated solves
QR	$A = QR$	Stable numerical computation

Table 1: Comparison of the principal methods.

There is therefore no universally "best" method. The appropriate algorithm depends on the structure of the matrix, numerical precision, number of right-hand sides, sparsity, and computational objective [5, 4].

14 Numerical Stability

Exact algebra and numerical computation are not identical.

A mathematically correct algorithm can be numerically poor if finite precision magnifies rounding errors. Gaussian elimination with partial pivoting is a standard practical strategy because it reduces the risk of using a very small pivot when a larger candidate is available [4, 5].

A related concept is the condition number. In a compatible matrix norm,

$$\kappa(A) = \|A\|\|A^{-1}\|.$$

A large $\kappa(A)$ indicates that small perturbations in the data can produce relatively large changes in the solution.

This distinction is essential:

$$\boxed{\text{algorithmic stability} \neq \text{problem conditioning.}}$$

A stable algorithm cannot make an intrinsically ill-conditioned problem well-conditioned [5].

15 Statistics

Linear systems occur naturally in statistical estimation.

In ordinary least squares, the model

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

leads, under the classical full-rank assumptions, to the normal equations

$$X^{\mathsf{T}}X\widehat{\boldsymbol{\beta}} = X^{\mathsf{T}}\mathbf{y}.$$

Thus estimating several regression coefficients requires solving a linear system. In modern numerical statistics, QR or singular-value decomposition is often preferred to directly solving normal equations because squaring the condition number can worsen numerical conditioning [20, 21].

16 Economics and Welfare Economics

Suppose three sectors produce quantities x, y, z, and three accounting constraints describe resource use:

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

This is structurally identical to the mathematical system introduced above.

Input–output economics provides a particularly important example. If A represents intermediate input coefficients and $\mathbf{d}$ represents final demand, the Leontief model is

$$\mathbf{x} = A\mathbf{x} + \mathbf{d}.$$

Hence

$$(I - A)\mathbf{x} = \mathbf{d},$$

and, when the inverse exists,

$$\mathbf{x} = (I - A)^{-1}\mathbf{d}.$$

The matrix $(I - A)^{-1}$ is the Leontief inverse and measures total direct and indirect production requirements [8].

In welfare economics, simultaneous equations can similarly represent resource constraints, market-clearing conditions, or equilibrium conditions. The linear system itself does not determine a welfare criterion; that requires an explicit social objective or welfare function [9].

17 Finance

Financial models frequently impose simultaneous constraints.

For example, suppose a portfolio contains three assets with quantities x, y, z. Three linear restrictions might be

$$\begin{aligned} x + y + z &= W, \\ r_1 x + r_2 y + r_3 z &= R, \\ \sigma_1 x + \sigma_2 y + \sigma_3 z &= S. \end{aligned}$$

If the coefficient matrix is nonsingular, the portfolio quantities are uniquely determined.

More sophisticated portfolio optimization is generally not merely a 3×3 linear system: covariance matrices, quadratic objectives, inequality constraints, and stochastic assumptions enter the problem. Nevertheless, linear algebra provides the computational foundation for many such models [10, 11].

18 Physics

Linear systems are ubiquitous in physics.

For example, equilibrium of three coupled quantities can produce

$$A\mathbf{x} = \mathbf{b}.$$

In mechanics, force-balance equations may be linearized around an equilibrium. In circuit theory, Kirchhoff's laws produce simultaneous linear equations for currents and voltages. In numerical physics, discretization of differential equations often produces large sparse linear systems [12].

A particularly simple conceptual example is

$$\begin{aligned} F_1 &= k_{11}x + k_{12}y + k_{13}z, \\ F_2 &= k_{21}x + k_{22}y + k_{23}z, \\ F_3 &= k_{31}x + k_{32}y + k_{33}z. \end{aligned}$$

This can be written as

$$\mathbf{F} = K\mathbf{x}.$$

The matrix K may represent a stiffness or response matrix.

19 Chemistry

Stoichiometric balancing can be formulated using linear algebra.

Suppose a chemical reaction contains several species. Let S be a stoichiometric matrix and $\mathbf{c}$ a vector of coefficients. Conservation requires

$$S\mathbf{c} = \mathbf{0}.$$

A nonzero vector in the null space of S represents a stoichiometrically balanced combination, subject to appropriate sign and normalization conditions [13].

Thus the mathematical operation is not restricted to numerical equations with arbitrary coefficients; it also describes conservation laws.

20 Biology

Compartment models, population models, metabolic networks, and gene-regulatory models can involve systems of linear equations.

For example, a linearized population model may be written

$$\mathbf{x}_{t+1} = A\mathbf{x}_t.$$

For a three-compartment model,

$$\begin{bmatrix} x_1(t+1) \\ x_2(t+1) \\ x_3(t+1) \end{bmatrix} = A \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}.$$

The eigenvalues of A then determine important aspects of long-run dynamics [14].

21 Psychology and Psychiatry

In psychology, linear algebra appears in psychometrics, factor models, covariance analysis, structural-equation models, and measurement theory.

A simplified latent-variable measurement model can be written

$$\mathbf{y} = \Lambda \boldsymbol{\eta} + \boldsymbol{\varepsilon}.$$

With three observed quantities and three latent or explanatory quantities, one obtains a structure resembling a 3×3 linear system.

The mathematical machinery should not, however, be confused with clinical diagnosis. Psychiatric phenomena are multidimensional, heterogeneous, and context-dependent. A linear model can represent a quantitative relationship without establishing a psychiatric cause, diagnosis, or treatment recommendation [15].

22 Philosophy of Modeling

A useful philosophical distinction is

$$\boxed{\text{mathematical validity} \neq \text{empirical validity.}}$$

If $A\mathbf{x} = \mathbf{b}$ is solved correctly, the mathematical conclusion is conditional upon the equations being an adequate representation of the problem.

The chain is therefore

$$\text{assumptions} \longrightarrow \text{model} \longrightarrow \text{equations} \longrightarrow \text{solution} \longrightarrow \text{interpretation.}$$

An exact solution to an incorrect model remains an incorrect description of the real-world phenomenon.

This is closely related to the philosophy of idealization: mathematical models deliberately omit some features of reality in order to isolate others [16].

23 Political Science

Political-science models can use simultaneous equations for voting, coalition formation, public-resource allocation, and institutional constraints.

For example, a stylized equilibrium model might be represented by

$$A\mathbf{x} = \mathbf{b},$$

where $\mathbf{x}$ contains policy or allocation variables and $\mathbf{b}$ contains exogenous constraints.

Such a model is only an abstraction. Political behavior involves strategic interaction, institutions, information, preferences, historical contingency, and power relations, many of which are not adequately represented by a small linear system [17].

24 Geopolitics, Diplomacy, and International Relations

A similar mathematical structure appears in abstract bargaining and resource-allocation models.

Suppose three parties have resource requirements represented by

$$A\mathbf{x} = \mathbf{b}.$$

A unique solution means that the specified linear constraints determine a unique allocation. Multiple solutions indicate degrees of freedom, while inconsistency means that the specified constraints cannot simultaneously be satisfied.

This interpretation is useful for understanding why negotiations often involve trade-offs: relaxing one constraint can enlarge the feasible set.

A simplified conceptual representation is

$$\mathcal{F} = \{\mathbf{x} : A\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \geq 0\}.$$

Diplomatic negotiation can then be viewed, in highly abstract form, as altering constraints or selecting among feasible outcomes. Actual international relations are substantially richer than such a model and include strategic behavior, uncertainty, institutions, credibility, and information asymmetry [18, 19].

25 Warfare Economics and Resource Allocation

Linear systems can also describe abstract resource-accounting problems associated with national security.

For example, suppose three broad resource categories must satisfy three aggregate requirements:

$$A \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}.$$

The variables can represent non-operational categories such as budgetary allocations, industrial capacity, or personnel totals.

The mathematical lesson is that simultaneous constraints can be solved by the same methods regardless of whether the variables represent economic quantities, physical quantities, or abstract resources.

This does not imply that real military planning can be reduced to a 3×3 linear system. Real systems involve uncertainty, nonlinear effects, adversarial adaptation, logistics, political constraints, and dynamic feedback.

26 Computer Science

Linear systems are fundamental to scientific computing.

The computational problem is

$$A\mathbf{x} = \mathbf{b}.$$

A generic solver can be represented by

```
function Solve3x3(A, b):
    validate dimensions
    determine whether A is singular or nearly singular

    if numerical solution is required:
        perform pivoted factorization
        solve triangular systems
    else:
        use exact arithmetic when appropriate

    verify residual r = b - A*x
    return x
```

The residual

$$\mathbf{r} = \mathbf{b} - A\mathbf{x}$$

is a useful diagnostic. A small residual demonstrates that the computed answer nearly satisfies the equations, although residual size alone does not fully characterize forward error when the system is ill-conditioned [5].

27 Algorithms and Complexity

For dense matrices of dimension n:

$$\text{Gaussian elimination} = O(n^3),$$

while solving the resulting triangular system is

$$O(n^2).$$

For $n = 3$, these asymptotic distinctions are practically insignificant. For large n, they become decisive.

If many right-hand sides share the same A,

$$A\mathbf{x}_i = \mathbf{b}_i,$$

factorization methods become especially attractive because the expensive factorization can be reused.

For sparse matrices, specialized sparse algorithms can be substantially more efficient than dense algorithms because they exploit zero structure [4, 7].

28 Data Science

Data-science pipelines often transform observations into matrices.

For a linear prediction model,

$$\widehat{\mathbf{y}} = X\boldsymbol{\beta},$$

estimating $\boldsymbol{\beta}$ involves linear algebra. A three-feature model can therefore contain a 3×3 system at the computational core.

Data preprocessing also matters. Scaling variables can improve numerical conditioning, while missing data, measurement error, confounding, and selection bias can invalidate conclusions even when the linear algebra is perfect.

Thus

$$\boxed{\text{good computation} + \text{bad data} \not\Rightarrow \text{good inference.}}$$

29 Artificial Intelligence and Machine Learning

Modern machine learning relies extensively on linear algebra.

A neural-network layer is fundamentally an affine transformation:

$$\mathbf{h} = W\mathbf{x} + \mathbf{b}.$$

Optimization algorithms repeatedly manipulate matrices, vectors, gradients, and Hessians.

A local quadratic approximation to a loss function has the form

$$L(\mathbf{x} + \Delta\mathbf{x}) \approx L(\mathbf{x}) + \nabla L(\mathbf{x})^{\mathsf{T}} \Delta\mathbf{x} + \frac{1}{2}\Delta\mathbf{x}^{\mathsf{T}} H \Delta\mathbf{x}.$$

Newton-type optimization consequently requires solving linear systems involving the Hessian:

$$H\Delta\mathbf{x} = -\nabla L.$$

Large-scale machine learning therefore depends on numerical linear algebra far beyond the simple 3×3 case [21].

30 A Unified Mathematical View

All of the methods considered can be interpreted as transformations of the same object:

$$A\mathbf{x} = \mathbf{b}.$$

The essential questions are:

1. Does a solution exist?
2. Is it unique?
3. How can it be computed?
4. How accurately can it be computed?
5. How sensitive is it to perturbations?
6. What does the solution mean in its application?

These questions correspond to different mathematical structures:

Question	Principal tool
Existence	Rank / consistency
Uniqueness	Determinant / rank / null space
Exact solution	Elimination / inverse / Cramer
Efficient repeated solution	LU factorization
Numerical stability	Pivoting / QR / SVD
Sensitivity	Condition number
Interpretation	Domain-specific model

Table 2: Comparison of questions and principal tools.

31 A Decision Procedure

For a practical 3×3 system, the following decision tree is useful.

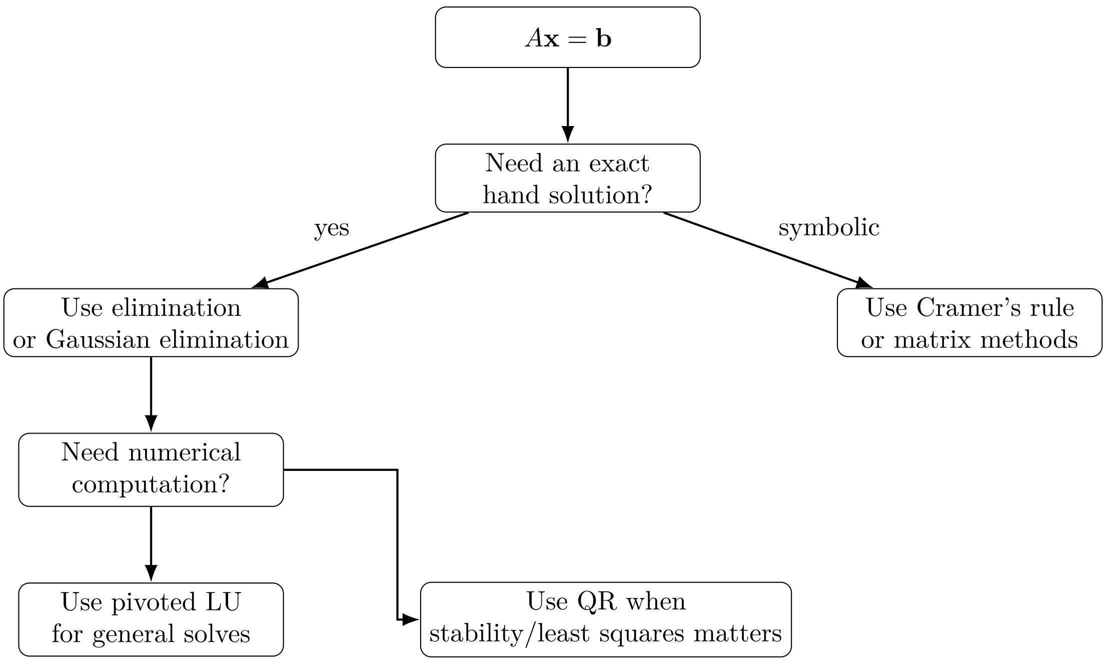

Figure 2: A decision procedure.

32 Worked Summary

For the system

$$\begin{aligned} x + y + z &= 6, \\ 2x - y + z &= 3, \\ x + 2y - z &= 2, \end{aligned}$$

the solution is

$$\boxed{x = 1, \qquad y = 2, \qquad z = 3}.$$

Verification gives

$$1 + 2 + 3 = 6,$$

$$2(1) - 2 + 3 = 3,$$

and

$$1 + 2(2) - 3 = 2.$$

Hence the candidate satisfies every equation.

Verification is an important final step in both symbolic and numerical problem solving.

33 Common Errors

1. Treating a singular matrix as invertible.
2. Dividing by a pivot without checking whether it is zero.
3. Losing a minus sign during elimination.
4. Confusing row operations with column operations.
5. Applying Cramer's rule when $D = 0$.
6. Computing an explicit inverse numerically when a factorization would be more appropriate.
7. Assuming a small residual automatically means a small forward error.
8. Treating a mathematical model as empirical proof.
9. Forgetting domain restrictions such as nonnegative quantities.

34 The Broader Lesson

The apparent simplicity of a 3×3 system conceals a central principle of applied mathematics:

$$\boxed{\text{many simultaneous relationships} \longrightarrow \text{linear representation} \longrightarrow \text{structured computation.}}$$

Substitution emphasizes algebraic dependence. Elimination emphasizes cancellation. Gaussian elimination emphasizes systematic row operations. Matrix inversion emphasizes operators. Cramer's rule emphasizes determinants. LU emphasizes factorization. QR emphasizes orthogonality and numerical stability.

These are not unrelated tricks. They are different perspectives on the same linear structure.

35 Conclusion

There are substantially more than three ways to solve a system of three linear equations in three unknowns.

The principal methods are:

substitution
elimination
Gaussian elimination
Gauss–Jordan elimination
matrix inversion
Cramer's rule
LU factorization
QR factorization

For ordinary hand calculations, elimination and Gaussian elimination are usually the most useful. Matrix inversion and Cramer's rule provide important conceptual and theoretical perspectives. LU factorization is highly useful for repeated numerical solves, while QR and related factorizations become increasingly important when numerical stability and least-squares problems matter.

Ultimately, the correct solution method is determined not only by the number of equations and unknowns but also by the purpose of the calculation, the structure and conditioning of the matrix, the desired precision, and the meaning assigned to the variables.

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Glossary

Augmented matrix
The matrix formed by adjoining the right-hand-side vector to the coefficient matrix, written $[A \mid \mathbf{b}]$.

Back-substitution
The process of solving a triangular system starting with the last equation and proceeding upward.

Condition number
A measure of sensitivity of a numerical problem to perturbations, commonly defined as $\kappa(A) = \|A\|\|A^{-1}\|$ for an invertible matrix.

Cramer's rule
A determinant-based formula for the components of the solution of a nonsingular square linear system.

Determinant
A scalar associated with a square matrix that indicates, among other things, whether the matrix is invertible.

Gaussian elimination
A sequence of elementary row operations used to transform a matrix into row-echelon form.

Gauss–Jordan elimination
An extension of Gaussian elimination that produces reduced row-echelon form.

LU factorization
A factorization $A = LU$ into lower- and upper-triangular matrices.

Null space
The set of vectors satisfying $A\mathbf{x} = \mathbf{0}$.

Pivot
A selected nonzero matrix entry used to eliminate other entries in its column.

QR factorization
A factorization $A = QR$ in which Q is orthogonal and R is upper triangular.

Rank
The dimension of the column space of a matrix, equivalently the number of linearly independent rows.

Residual
For an approximate solution $\widehat{\mathbf{x}}$, the vector $\mathbf{r} = \mathbf{b} - A\widehat{\mathbf{x}}$.

Singular matrix
A square matrix with determinant zero, and therefore no inverse.

Triangular system
A system whose coefficient matrix is triangular and can therefore be solved efficiently by substitution.

The End